# QUBO Formulation: Quantum-Circuit Qubit Allocation

## Quantum-Circuit Qubit Allocation

**Problem Statement.** Given a set of logical qubits $L$, a set of physical qubits $P$, and a hardware coupling graph characterized by pairwise penalties $W_{p,q}$ for mapping logical interactions to physical pairs $(p, q)$, the objective is to determine a one-to-one assignment of logical qubits to physical qubits. The assignment must minimize the total penalty incurred by two-qubit gates, where $W_{p,q}$ typically encodes hardware distance or inverse gate fidelity. If $|L| > |P|$, the problem is infeasible.

**Decision Variables.** For each logical qubit $l \in L$ and physical qubit $p \in P$, introduce a binary variable $x_{l,p} \in \{0, 1\}$, where $x_{l,p} = 1$ if logical qubit $l$ is mapped to physical qubit $p$, and $x_{l,p} = 0$ otherwise.

**QUBO Derivation. Step 1 – Objective Function.** The original optimization problem minimizes the total allocation cost:

$$\min_{x} \quad C(x) = \sum_{(i,j)\in E_{\log}} \sum_{p\in P} \sum_{q\in P} W_{p,q}\, x_{i,p}\, x_{j,q}, \tag{1}$$

where $E_{\log}$ denotes the set of logical two-qubit gate edges. Diagonal terms $W_{p,p}$ are set to a sufficiently large constant to discourage self-mapping.

**Step 2 – Constraints.** The assignment must satisfy two sets of constraints:

$$\sum_{p\in P} x_{l,p} = 1 \quad \forall l \in L, \tag{C1}$$

$$\sum_{l\in L} x_{l,p} \leq 1 \quad \forall p \in P. \tag{C2}$$

**Step 3 – Penalty Terms.** We convert the constraints into quadratic penalty terms using a positive weight $\lambda$.

For (C1), we use the standard squared penalty:

$$P_1(x) = \lambda \sum_{l\in L} \left( \sum_{p\in P} x_{l,p} - 1 \right)^2. \tag{2}$$

Expanding the square and applying the idempotent property $x_{l,p}^2 = x_{l,p}$ yields:

$$\begin{aligned} P_1(x) &= \lambda \sum_{l \in L} \left( \sum_{p \in P} x_{l,p}^2 + 2 \sum_{p<q} x_{l,p} x_{l,q} - 2 \sum_{p \in P} x_{l,p} + 1 \right) \\ &= \lambda \sum_{l \in L} \left( 1 - \sum_{p \in P} x_{l,p} + 2 \sum_{p<q} x_{l,p} x_{l,q} \right). \end{aligned} \tag{3}$$

For (C2), we use the inequality penalty form $\lambda \sum_p (\sum_l x_{l,p})(\sum_l x_{l,p} - 1)$, which naturally enforces the $\leq 1$ condition without linear or constant terms:

$$P_2(x) = \lambda \sum_{p \in P} \left( \sum_{l \in L} x_{l,p} \right) \left( \sum_{l \in L} x_{l,p} - 1 \right) = \lambda \sum_{p \in P} \sum_{l \neq l'} x_{l,p} x_{l',p}. \tag{4}$$

**Step 4 – Combined Hamiltonian.** The full QUBO Hamiltonian $H(x)$ is the sum of the objective and penalty terms:

$$H(x) = C(x) + P_1(x) + P_2(x). \tag{5}$$

**Step 5 – Q Matrix and Offset.** Reading off the coefficients from $H(x)$, the QUBO matrix entries $Q_{(l,p),(l',p')}$ and constant offset $c$ are given by:

$$Q_{(l,p),(l,p)} = -\lambda, \tag{6}$$
$$Q_{(l,p),(l,q)} = 2\lambda \quad \text{for } p \neq q, \tag{7}$$
$$Q_{(l,p),(l',p)} = \lambda \quad \text{for } l \neq l', \tag{8}$$
$$Q_{(i,p),(j,q)} = W_{p,q} \quad \text{for } (i,j) \in E_{\log}, \tag{9}$$
$$c = \lambda |L|. \tag{10}$$

All other entries are zero.

**Penalty Weight Justification.** Let $W_{\max} = \max_{p,q} |W_{p,q}|$ and let $d_{\max}$ be the maximum degree of any logical qubit in $E_{\log}$. Violating a single constraint can at best reduce the objective cost by an amount bounded by $d_{\max} W_{\max}$. To guarantee that any infeasible assignment has strictly higher energy than the optimal feasible assignment, the penalty weight must satisfy:

$$\lambda > d_{\max} \cdot W_{\max}. \tag{11}$$

In practice, a conservative choice such as $\lambda > 2 \max_{p,q} |W_{p,q}|$ or $\lambda = 100$ (for normalized $W_{p,q} \in [0,1]$) is sufficient to dominate any feasible objective improvement.

**Correctness Argument.** Minimizing $H(x)$ over $\{0,1\}^{|L|\times|P|}$ recovers the optimal allocation because $H(x) = C(x)$ for all feasible $x$, and $H(x) > C(x)$ for all infeasible $x$. Specifically, any violation of (C1) or (C2) introduces a penalty term strictly greater than $\lambda$. Since $\lambda$ is chosen to exceed the maximum possible reduction in $C(x)$ achievable by violating constraints, the global minimum of $H(x)$ cannot occur at an infeasible point. Consequently, the ground state of $H(x)$ corresponds exactly to a feasible mapping that minimizes the original allocation cost.

**Illustrative Example.** Consider a small instance with $|L| = 3$ logical qubits, $|P| = 3$ physical qubits, and logical edges $E_{\log} = \{(0,1),(0,2),(1,2)\}$. Let all hardware penalties be uniform, $W_{p,q} = 1$ for $p \neq q$, and $W_{p,p} = 10$. Set $\lambda = 100$. The objective becomes:

$$C(x) = \sum_{(i,j)\in\{(0,1),(0,2),(1,2)\}} \sum_{p=0}^{2}\sum_{q=0}^{2} W_{p,q}\, x_{i,p}x_{j,q}. \tag{12}$$

The penalty terms expand to:

$$P_1(x) = 100\sum_{l=0}^{2}\left(1 - \sum_{p=0}^{2} x_{l,p} + 2\sum_{0\le p<q\le 2} x_{l,p}x_{l,q}\right), \tag{13}$$

$$P_2(x) = 100\sum_{p=0}^{2}\sum_{0\le l<l'\le 2} x_{l,p}x_{l',p}. \tag{14}$$

The full Hamiltonian $H(x) = C(x) + P_1(x) + P_2(x)$ contains $3 \times 3 = 9$ binary variables. The constant offset is $c = 100 \times 3 = 300$. The optimal solution maps logical qubits $\{0,1,2\}$ to physical qubits $\{0,1,2\}$ in a permutation that aligns logical edges with the lowest physical penalties (e.g., $x_{0,0} = x_{1,1} = x_{2,2} = 1$, yielding $C(x) = 3$ and total energy $H(x) = 303$). Any deviation from a valid permutation incurs a penalty of at least 100, strictly increasing the energy.